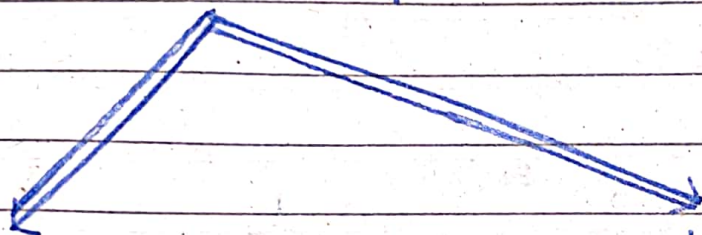


Wave & Oscillation :-

Free Oscillation



Restoring Force $\Rightarrow$ Oscillation for infinite time period with ext. force



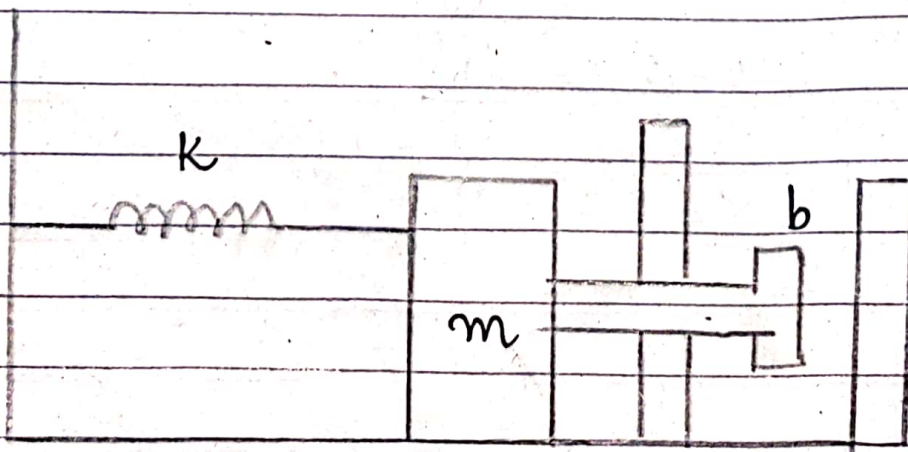
Damped

friction force $\neq 0$
(Amplitude decrease with time)

Undamped

friction force = 0
(Amplitude Constant remain same)

* Damped Harmonic Motion:



(18)

friction

If the or resistive forces are present, then the amplitude of oscillation gradually decreases to zero. As a result of friction, the motion is called to be damped by friction.

Often the friction arises from air resistance or internal force and is called damped Harmonic Motion.

In most cases, frictional force is proportional to the velocity of body but directed opposite to it.

Consider a Harmonic oscillation the net force (restoring force) is " $-kx$ " & damping force (drag force) is " $-bv$ ", where " b " is a +ve constant & depends upon the properties of fluids. Such as its density and on the shape or dimension of object.

From Newton's Second law of Motion:

$$\Sigma F = ma$$

$$-kx - bv = ma$$

$$-kx - b \frac{dx}{dt} = m \frac{d^2x}{dt^2}$$

$$\frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

$$\frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \omega^2 x = 0$$

$\frac{b}{m}$

represent damping factor

This equation is a second order differential eq. A solution of this eq will be,

$$\therefore \frac{d^2 x}{dt^2} + \omega^2 x = 0$$

$$x(t) = x_m \cos(\omega t + \phi)$$

$$x(t) = x_m \sin(\omega t + \phi)$$

$$x = x_m e^{-bt/2m} \cos(\omega' t + \phi)$$

where,

$$\omega' = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$$

This equation shows that amplitude of motion is exponentially decreasing.

A

$$t = T = \frac{2\pi m}{b}$$

Time Period:

$$\text{Time period} = \frac{2\pi}{\omega}$$

$$T = \frac{2\pi}{\sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}}$$

Frequency:

$$\text{Frequency} = \frac{\omega'}{2\pi}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{\kappa}{m} - \left(\frac{b}{m}\right)^2}$$

$$F = -Kx - bVx$$

$$F = m a_x$$

$$m a_x = -Kx - bVx$$

$$m \frac{d^2 x}{dt^2} = -Kx - b \frac{dx}{dt} \quad \text{--- (A)}$$

By simplification we get:

$$\left[x = A e^{-\frac{b}{2m}t} \cos(\omega t + \phi) \right] \quad \text{--- (B)}$$

and angular frequency

is

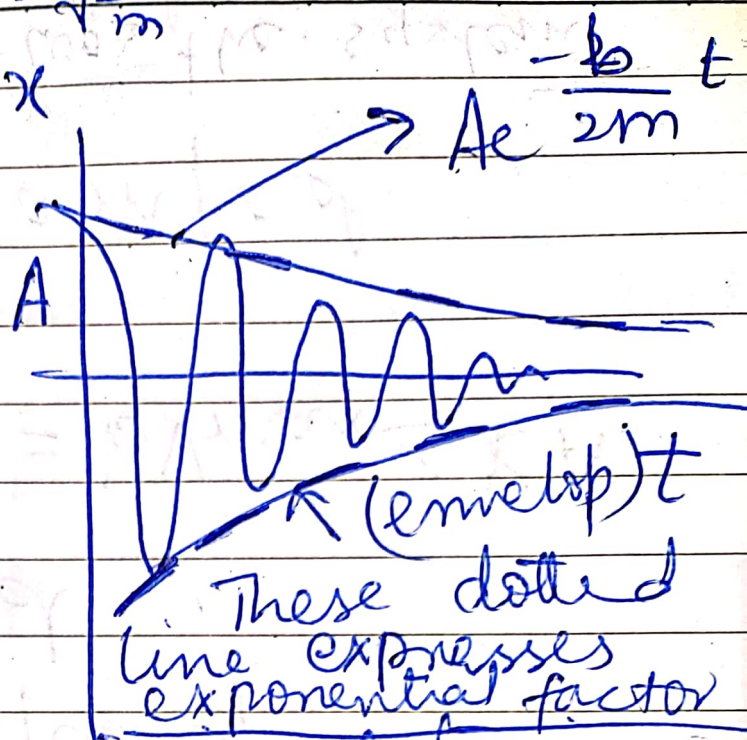
$$\omega^2 = \sqrt{\frac{K}{m} - \left(\frac{b}{2m}\right)^2} \quad \text{--- (C)}$$

$$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

Date / /

9) you substitute x in (A) you can verify (C)



Graph shows the position as a function of time for an object oscillating in the presence of retarding force.

We see that when the retarding force is small, the oscillatory character of the motion is preserved but the amplitude decreases in time, and the motion ultimately stops. Any system that behaves in this way is damped oscillator.

where $\omega_0 = \sqrt{k/m}$

represents the angular frequency in the absence of retarding force (undamped oscillator) and is called natural frequency.

There are three different cases of damping.

① Underdamped

when $R_{\max} = bV_{\max} < KA$

② Critically damped

when $b/2m = \omega_0$ the system does not oscillate and called C.D.

② Critically damped

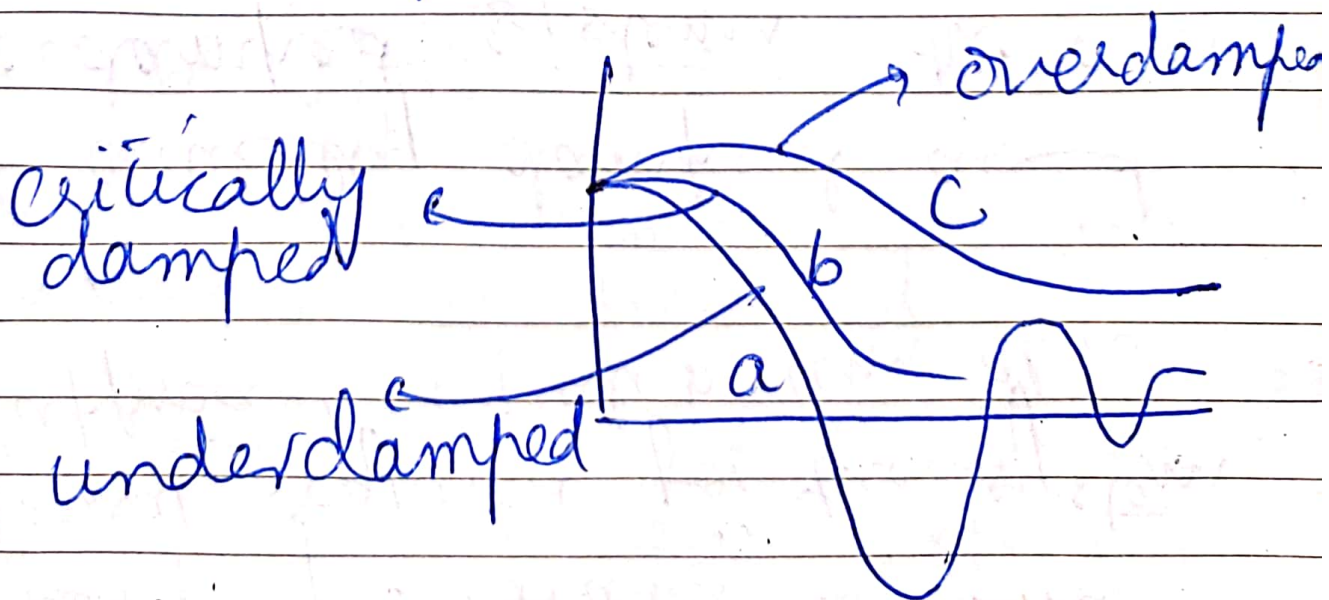
when $b/2m = \omega_0$ the system does not oscillate and called C.D.

③ Overdamped:

If the medium is so viscous that retarding force is greater than restoring force.

$$R_{\max} = bV_{\max} > KA$$

and $\frac{b}{2m} > \omega_0$ the system is overdamped.



As damping increases the time interval required for the system to approach equilibrium also increases.

For critically damped and overdamped systems there is no angular frequency ω and the solution is not valid of eqn (C).

Whenever friction is present in a system, whether the system is overdamped or underdamped, the energy of the oscillator eventually falls to zero.